

○ Circled edges indicates detection timing.

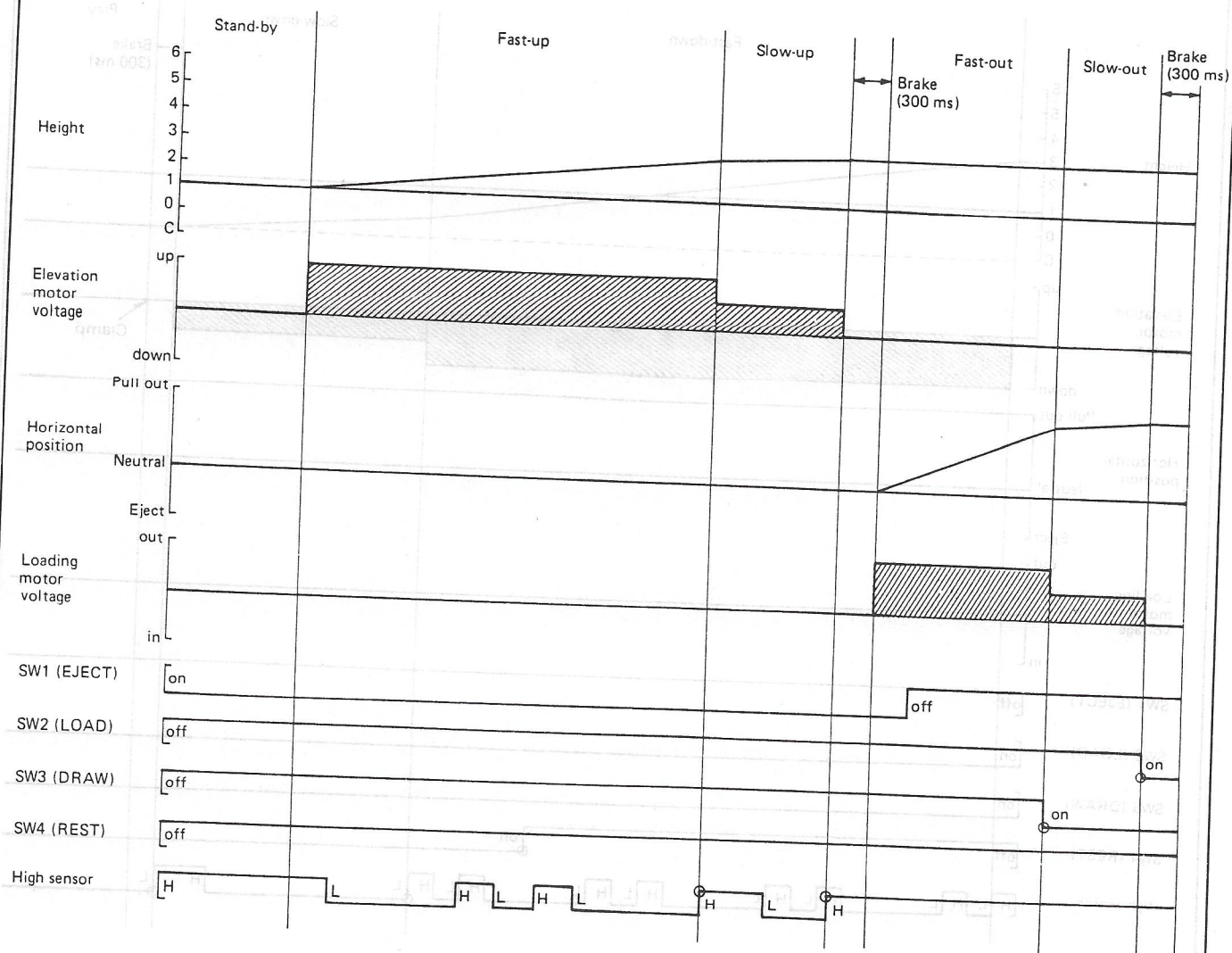


Figure 21-14 Timing Chart

(2) Open position

Clamp

○ Circled edges indicates detection timing.

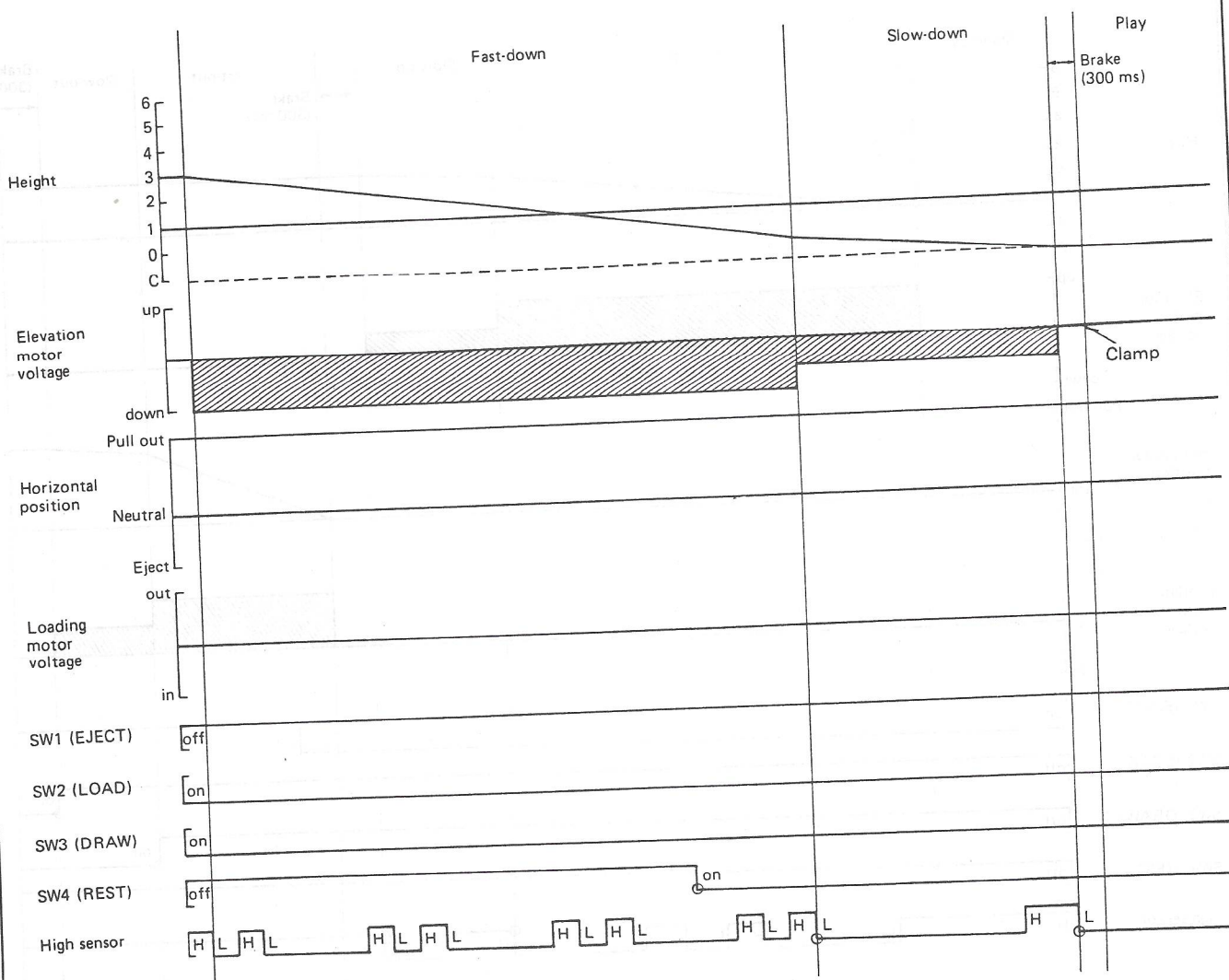


Figure 21-15 Timing Chart



○ Circled edges indicates detection timing.

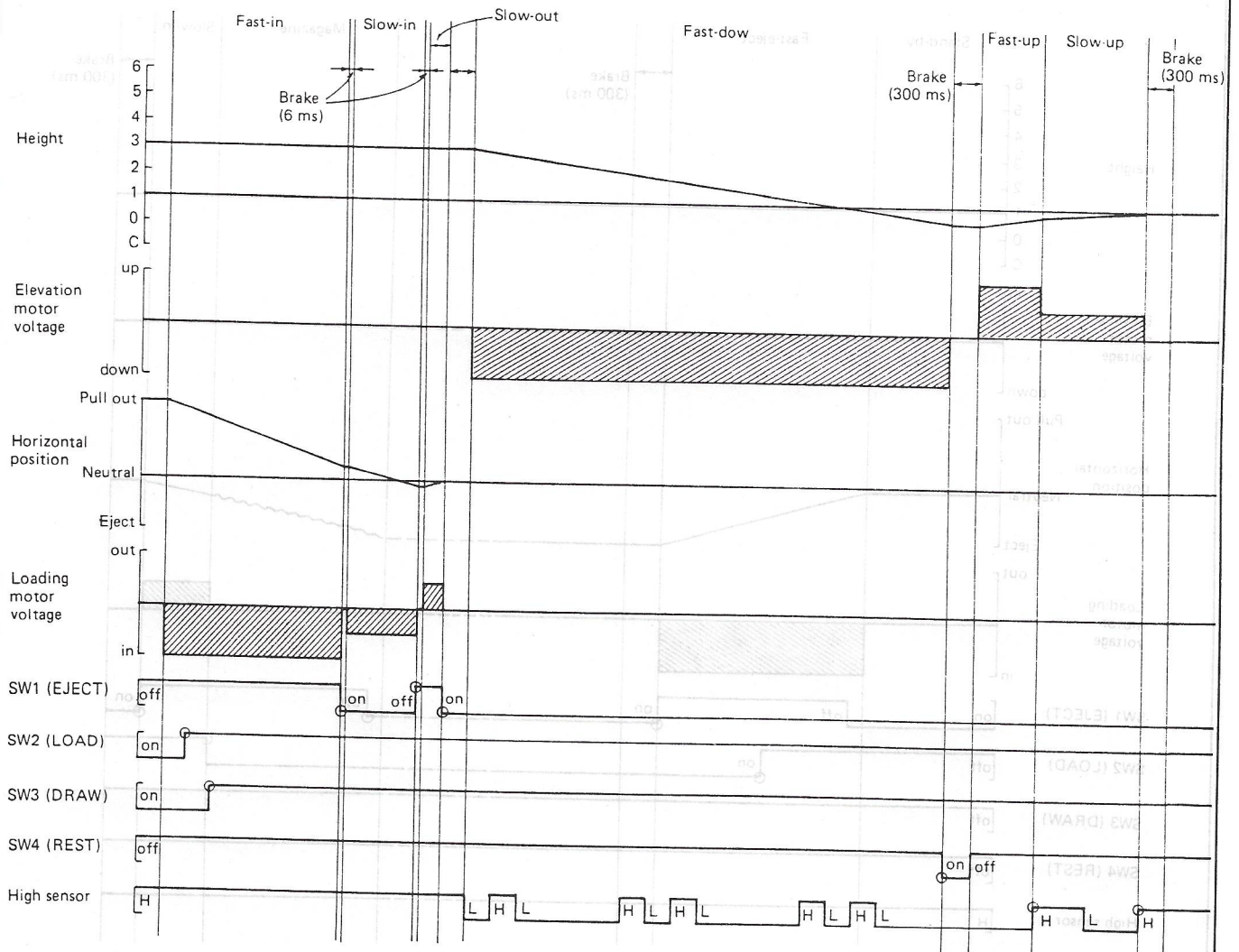


Figure 21-16 Timing Chart

(4)

Standby

Eject

Magazine Set

○ Circled edges indicates detection timing.

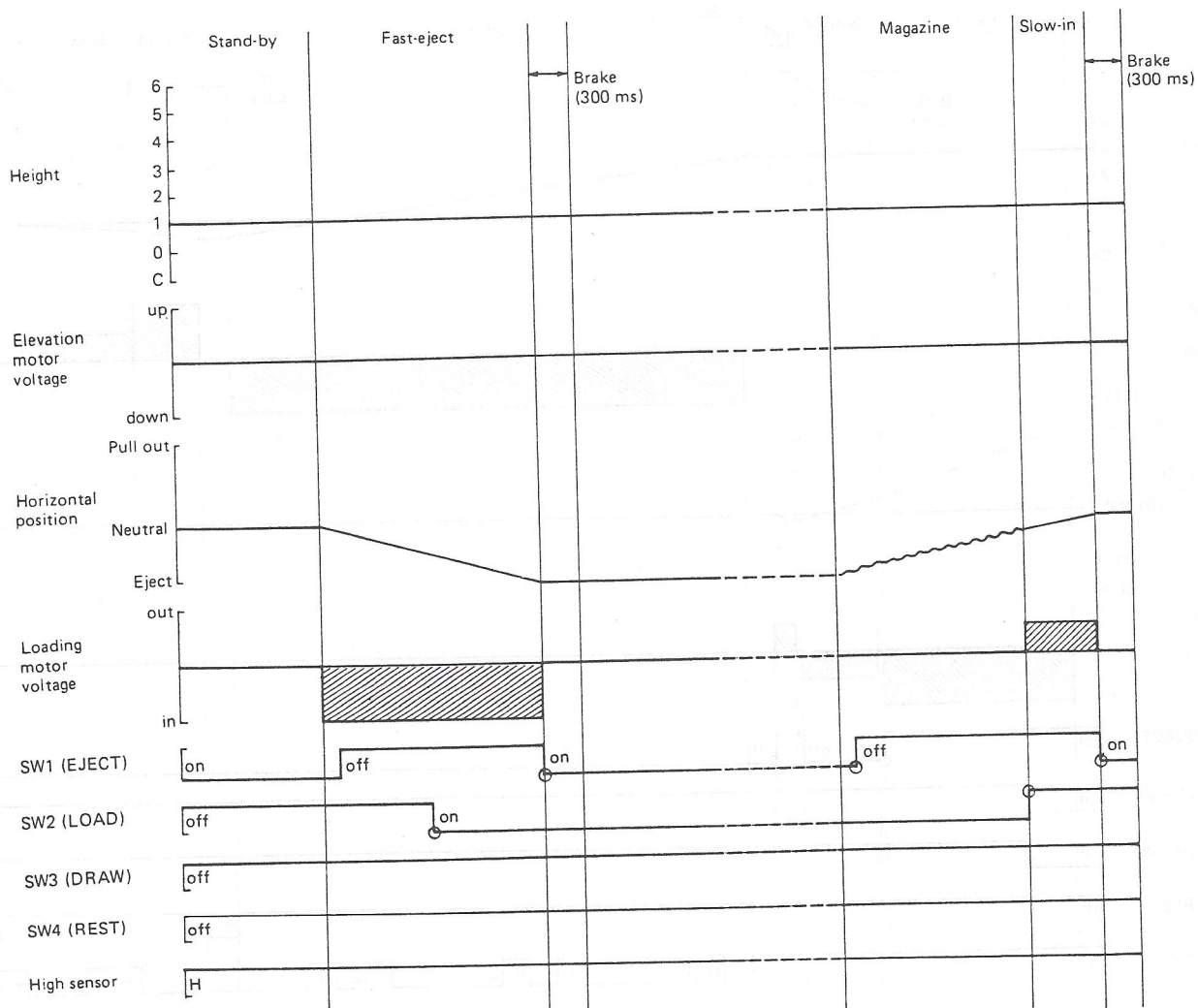


Figure 21-17 Timing Chart

5 Problem Solving

Symptom	Possible cause	Check point
At power on, the disc loader is lowered to the lowest position and remains there.	Rest switch is defective. Elevation motor has been stopped by an ON signal from the rest switch.	Check the Rest Position Sense switch (S454) for correct ON state.
At power on, the disc loader continues elevation for about 10 seconds.	Height sensor signal is not applied to the CPU. The CPU cannot compute disc loader height from the sensor pulses.	Check the sensor (D451) output while the elevation motor is rotating.
Disc magazine is ejected soon after it is inserted.	Magazine switch remains off.	Check the Magazine switch (S462) for proper ON state.
Disc tray is pulled out twice and "No Disc" indication appears. When the Play switch is pressed, message "-E" appears.	Draw switch remained off when the disc tray was pulled out. The CPU identifies it as no disc and returns the tray.	Press the Draw switch (S453) and check for correct ON state.
The same disc is pulled out then returned repeatedly at about 10 second interval.	(1) Mechanism operation is impeded in the pulling process. (2) Loading switch remains off at the fully pulled position.	(1) Manually rotate the loading motor to check for smooth mechanism movement. (2) Check the Loading switch (S452) for correct ON state.
When the magazine is ejected, the player displays loading indication for about 10 sec., then returns to the normal state.	(1) Mechanism operation is impeded in the eject process. (2) Eject or Loading switch remains off when the magazine is ejected.	(1) Manually rotate the loading motor to check for smooth mechanism movement. (2) Check the Eject switch (S451) and Loading switch (S452) for correct ON state.
At power on, all the display digits and indicators momentarily come on, then immediately go to the loading message. However, key entry is disabled after mechanism reset operation is completed.	Any of the key switches is left closed, causing CPU runaway.	Check for defective key switch.
"No Disc" message appears 3 seconds after a disc is clamped, then the disc is returned to the magazine.	Optical pickup is unfocused.	Check for optical deck misalignment. Rest position height error or disc clamp error due to height sensor signal error.

4. Height Adjustment of the Elevator Unit

At rest (standby) position, the elevator unit must be positioned 1.6mm above the magazine holder surface.

The purpose of this adjustment is to set this relative position. When performing this adjustment, the shutter is adjusted so that the pulse count (generated by the photo detector and detected by the CPU) corresponds to the proper height. If the height is not properly set, the CPU does not detect the proper reference, causing the elevator unit to either miss the proper disc tray, access two disc trays, or access the wrong disc tray.

PROCEDURE

- (1). Remove the metal cover plate covering the magazine holder compartment.
- (2). Switch the power to "ON"
- (3). The left, front edge of the metal cover plate is provided with a 1.6mm gauge.

Rest the gauge side of the plate on the right side surface of the magazine holder. Confirm the distance between the front edge of the elevator unit's chassis surface and the surface of the magazine holder.
- (4). If the distance is not within 1.6 $\pm$ 0.2mm, adjust the height by turning the HEIGHT ADJUSTMENT screw located on the top side of the ELEVATOR GEAR shows figure 21-18.
- (5). Insert the magazine in its compartment, and select the sixth disc tray, via the front panel. If the proper tray is not accessed, repeat steps 3 and 4 above.

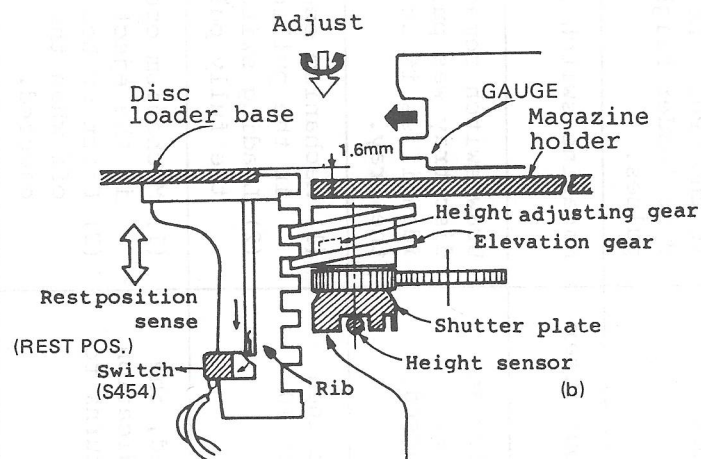


Figure 21-18 Positioning of the gauge

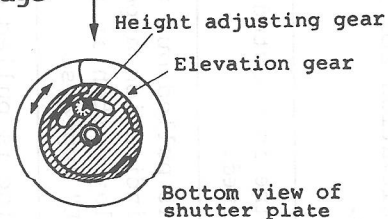


Figure 21-18 Top view of the elevator gear

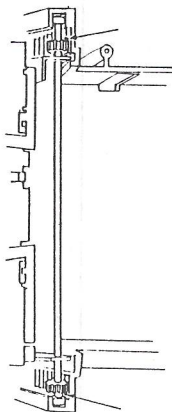
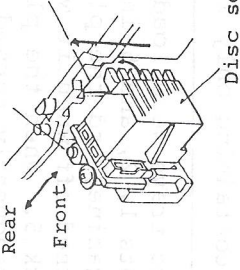
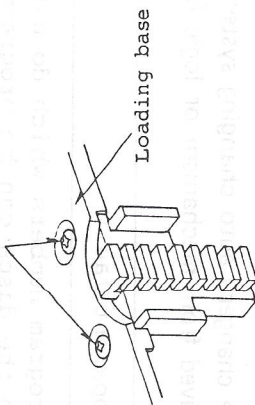
6 Apparent Fault

The changer's auto changing system may exhibit the following malfunctions but they may not be derived from mechanism or logic failure:

Apparent fault	Possible cause
1 Program numbers which do not exist on the discs can be programmed in the Program mode but nothing is played in Play mode. (e.g. the 25th track was programmed for a CD containing only 12 tracks.)	Other CD players first read the TOC just after a disc is loaded, then enter the Pause mode to wait for the next command. They can verify the validity of the programmed contents by comparing the number of tracks on the disc with that in the program. However, the SL-P600C takes so much time to read the TOCs on all of the six discs that it ignores invalid track numbers in the program.
2 Disc trays were loaded only in slots 1, 2, and 5 in the disc magazine, and the player was turned off when it was playing Disk 5. When the player was subsequently turned on, Disk 5 was returned to the tray for Disk 3 (disk position has changed in the magazine).	As long as the player is left turned on, the CPU memorizes which slot the disc now on the playing deck came from. However, once the player has been turned off, the CPU loses its memory and cannot determine to which slot the disc should be returned. In such a case, the CPU sequentially searches for a vacant slot from bottom to top, and returns the disc into the first vacant slot found. At an occupied slot, the Load switch remains off. The CPU checks the time until the Load switch is turned on after a return sequence was started (Draw switch S453 was turned off). If it is too long (more than ms), the CPU identifies that the slot is occupied. * This problem can be avoided if all of the six trays are always loaded in the magazine (discs need not be loaded in all of the six slots), or if trays are loaded from slot 1 with no intermediate vacant slot left in the magazine.

7 Mechanism Check Points for Servicing

When disassembling or reassembling the deck mechanism, observe the following cautions:

Item	Caution	Symptom caused by misalignment
1 Magazine base assembly (* Sidewise gear deviation) 	The magazine base gear should have no sidewise deviation from the timing gear on the loading base. Timing gear (right) Magazine base Check the sidewise deviation of the gear on the magazine base side, by using the projection on the gear as a guideline.	Tray open/close operation failure.
2 Disc sensor switch assembly (S455-S460) 	If the disc sensor switches are skewed, they may fail to sense the discs. Swing shaft A in the longitudinal direction to the following design height. Design height: 28.15 to 27.75 mm Magazine holder base Disc sensor switch	Disc sensor failure
3 Elevation cam retention Retention screws 	If the retention screws are loose, malfunction or mechanism noise may occur due to insufficient elevation stroke. Be sure to tighten them firmly.	Disc tray operation failure Mechanism noise.

8 Pin Description of MN15283PJB

• MN15283PER (System Control)

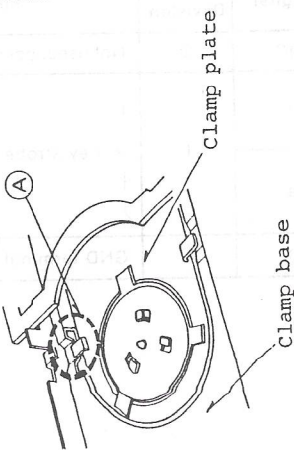
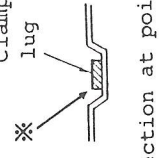
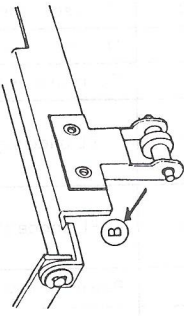
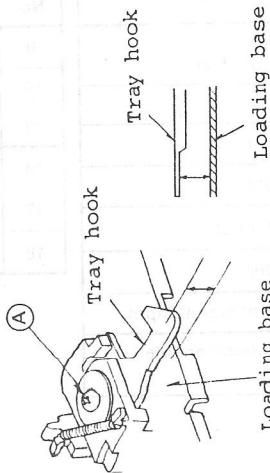
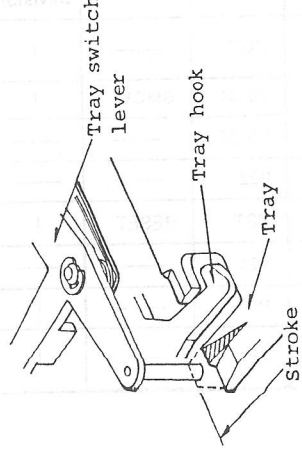
Pin No.	Mark	I/O	Function
1	GND	I	GND terminal
2	XO	O	Not used, open
3	X1	I	Optical servo condition input
4	M-RLY	I	Muting control
5	MDATA	O	Command data output
6	MCLK	O	Command clock output
7	MLD	O	Command load output
8	SLOW	O	Motor drive control
9	FOW/REV	O	Motor drive control
10	LOADING	O	Motor drive control
11	HIGHT	O	Motor drive control
12	SYNC	I	Not used, open
13	RST	I	Reset signal input (reset at "L")
14	BLKCK	I	Sub-code block (Q-data) clock (75kHz) input
15 ~ 18	P50 ~ P53	I	Key scan input
19	CLOCK	I	Sub-code frame clock (7.35kHz)
20	SUBQ	I	Sub-code "Q" data input
21	—	—	Not used, open
22	—	—	Not used, open
23	TRV-R	O	Traverse reverse command signal
24	TRV-F	O	Traverse forward command signal
25	CONT4	O	Optical servo IC control signal (KIKCR: Kick direction [Reverse] command)

Pin No.	Mark	I/O	Function
26	CONT3	O	Optical servo IC control signal (KICKF: Kick direction [Forward] command)
27	CONT2	O	Optical servo IC control signal (TRON: Tracking servo)
28	CONT1	O	Optical servo IC control signal (FOON: Focus servo)
29	STAT	I	Key input strobe and processing status input from signal processing LSI travers position detector
30	HIGHT	I	Elevator height signal input
31	FLOCK	I	Optical servo condition (Focus) input
32	SENSE	I	Optical servo condition (Track cross) input
33	PAUSE	O	Not used, open
34	PLAY	O	Not used, open
35	DAC	O	Not used, open
36	VPP	I	FL drive power supply (connected to -32V)
37 ~ 52	DO ~ DF	O	FL glide signal and key scan signal
53 ~ 61	S8 ~ S0	O	FL anode signal
62	VDD	I	Power supply (connected to +5V)
63	OSC2	I	Clock terminal
64	OSC1	I	Clock input

• MN1550PEB (Remote Control Signal Processing)

Pin No.	Mark	Signal	I/O Devision	Function
1	VDD	—	I	Power supply (connected to +5V)
2	OSC1	SMCK	I	Clock input
3	OSC2	—	—	Not used, open
4	P22	—	—	Not used, open
5	RST	RESET	I	Reset signal input
6	P21	—	—	Not used, open
7	P20	—	I	Remote control signal input
8	IRQ	—	I	Program enable/de-enable control

Pin No.	Mark	Signal	I/O Devision	Function
9	P31	SYNC	O	Not used, open
10 } 13	P10 } P13	Data	I	Key strobe
14 } 17	P00 } P03	Data		
18	VSS	GND	I	GND terminal

4		When the mechanism is lowered (in the Play position), the clamp plate "lug" must rest in the recess of the clamp base (marked with an asterisk). Clamp plate lug ※ Cross section at point A If there is any clearance between the two, tilt the spring on the clamp plate in the direction of arrow (B) to increase the stroke.  Diagonal rear view of clamp plate 	Disc does not rotate. Mechanism noise heard during disc rotation.
5	Unstable tray open operation 	Major causes will include: 1) Tray hook skew. 2) Loading base height defect. Verify that the distance between the top surface of the tray hook and that of loading base is 4.3 mm. If necessary, adjust with (A) or replace the tray hook (SLD75).	Disc tray fails to close.
6	Unstable tray loading (draw switch actuation error) 	The tray switch lever activation stroke is mechanically limited. So it is likely that draw switch (S453) activation failure will be due to the tray switch end pins failing to catch the tray edge (deformed tray switch lever). Action: Replace tray hook, SLD71.	Failing to enter Play mode (the mechanism attempts tray close drive twice for all of six disc trays, and finally stops in the neutral state).

22. Reference

1. Mechanism Adjustment

When the optical pickup and peripheral mechanism parts incorporated in the traverse deck are replaced or removed, such as the spindle motor, it is necessary to confirm the adjustment related to the optical pickup and then readjust as required.

A. Turntable Height Adjustment

a) Purpose

The output from the laser diode is focused on the signal side of the disc, via the focus lens. Focussing is

accomplished by moving the focus lens vertically. The focus servo is used to control such vertical movement.

Since the focus servo is designed to operate within a limited range, it is necessary to adjust the lens at the mechanical midpoint position of the servo's respective operating range. If this adjustment should be deviated, skipping may occur during play mode or in an extreme case, no turntable rotation, etc.

b) Positional relation between optical pickup and disc surface

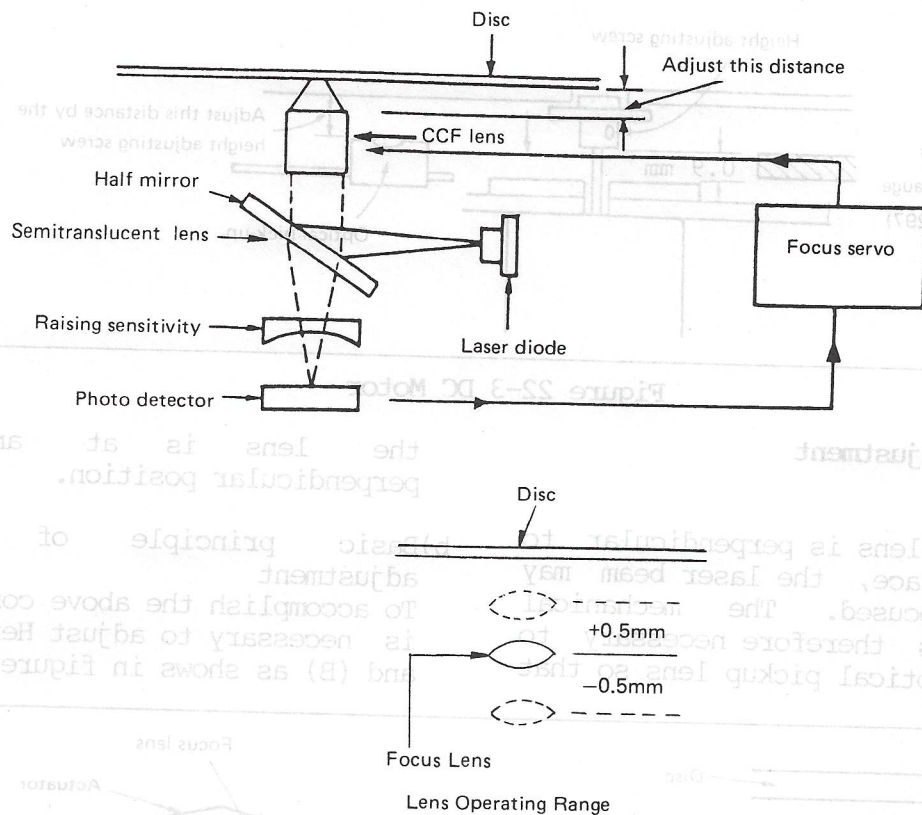


Figure 22-1 Beam Method

c) Basic principle of height adjustment of turntable height

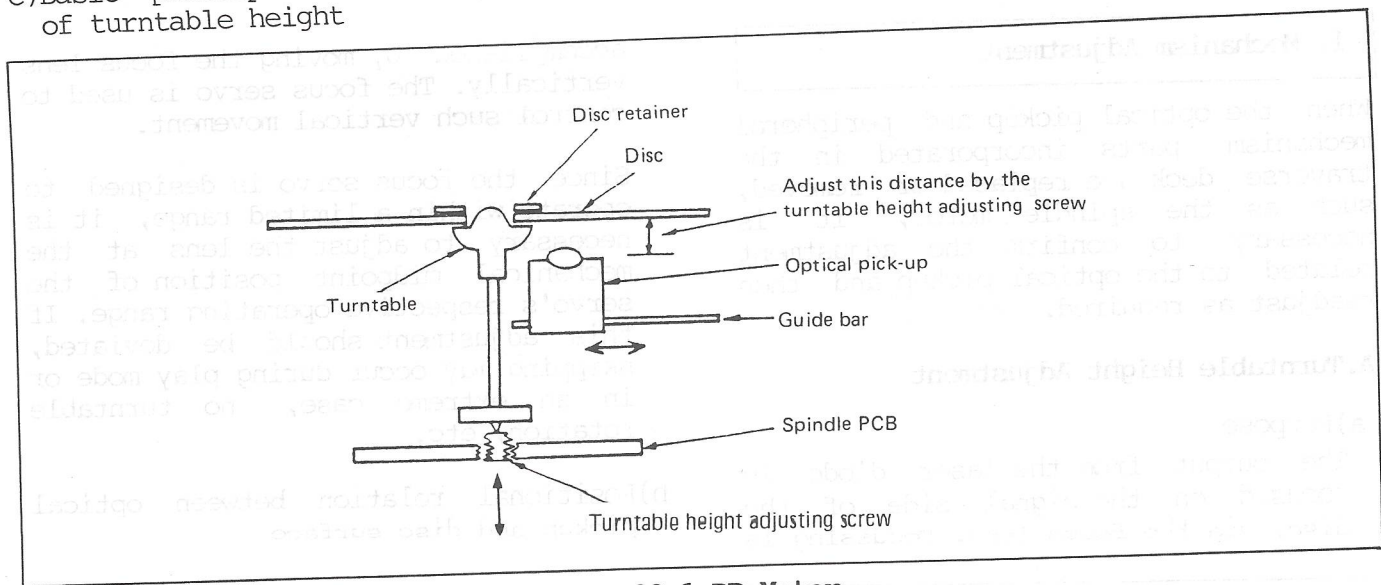


Figure 22-1 DD Motor

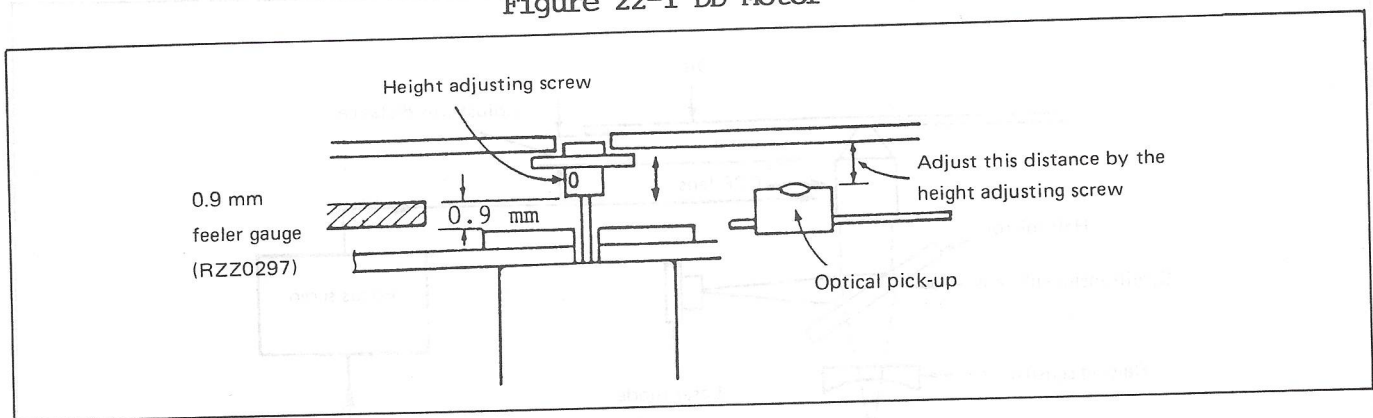


Figure 22-3 DC Motor

B. Mechanical Adjustment

a) Purpose

Unless the lens is perpendicular to the disc surface, the laser beam may not be focused. The mechanical alignment is therefore necessary to adjust the optical pickup lens so that

the lens is at an optimum perpendicular position.

b) Basic principle of mechanical adjustment

To accomplish the above condition, it is necessary to adjust Hex screws (A) and (B) as shows in figure 22-4.

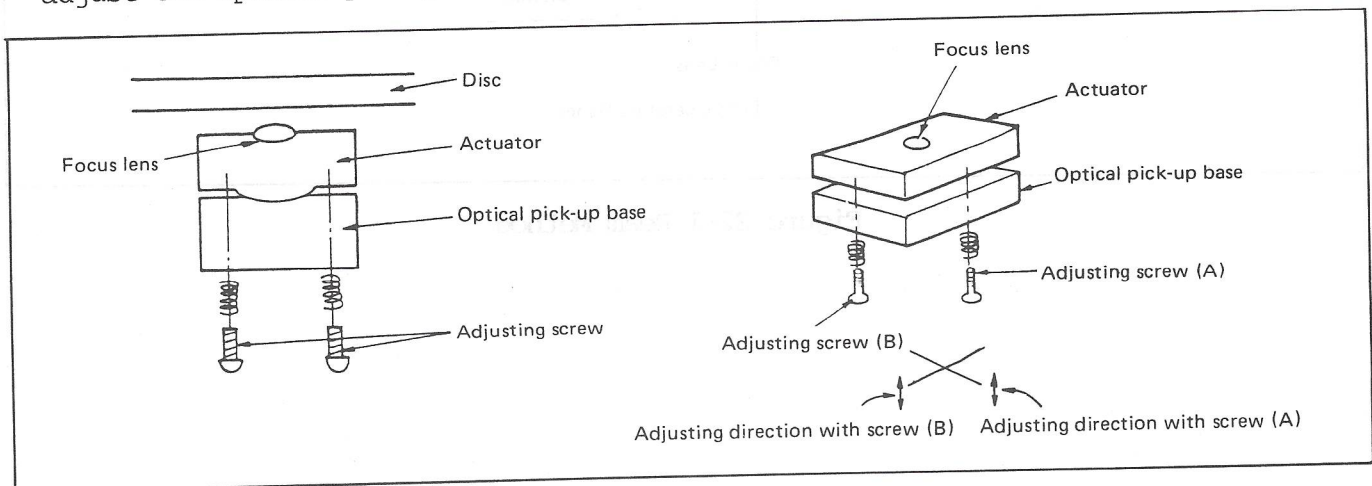


Figure 22-4

2. Electrical Adjustment

These adjustments are so important to the extent to the performance of the CD player depends mainly on their accuracy.

A. Best Eye (PD Balance) Adjustment

a) Purpose

Due to fluctuations of as described earlier, under "turntable height adjustment" the focus servo is designed to work within a certain

operating range. The purpose of this adjustment is to set the focus servo's reference voltage (zero); at which point the lens focusing depth is set.

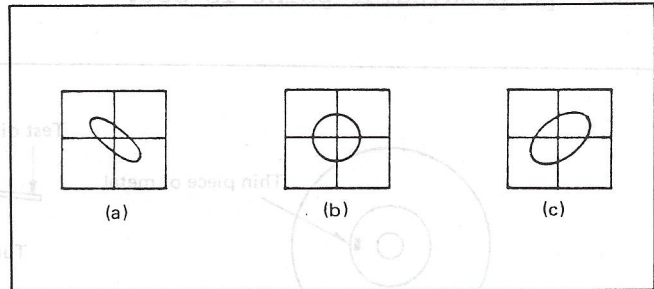


Figure 22-8 Beam Shapes Received by the Photodetector

b) Basic principle of best eye adjustment and best point

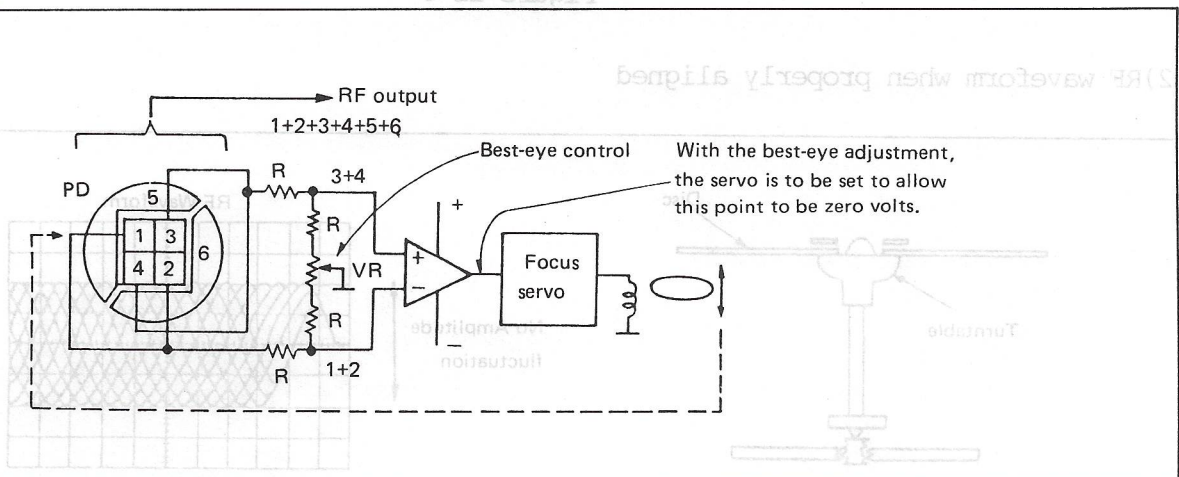
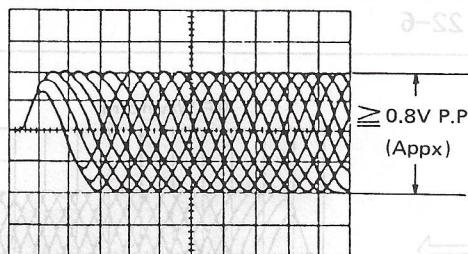


Figure 22-9 Focus Servo Loop



Maximum amplitude of the RF waveform is achieved when the best eye adjustment is properly set, as described above.

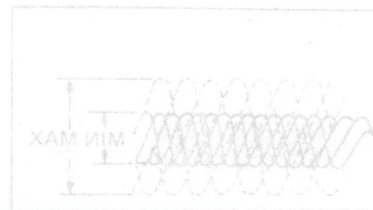


Figure 22-10 RF Waveform

c) Basic principle of adjustment method

The warped TEST DISC (SZZP1056C) is used to simulate the non-perpendicular condition. While playing the disc, the RF amplitude's vertical fluctuation is to be minimized; at which point the most perpendicular point is set.

(1) RF waveform at use of the TEST DISC (SZZP1056C)

A thin piece of metal is mounted to the part where the disc touches the turntable. The RF waveform is fluctuated vertically (amplitude fluctuated).

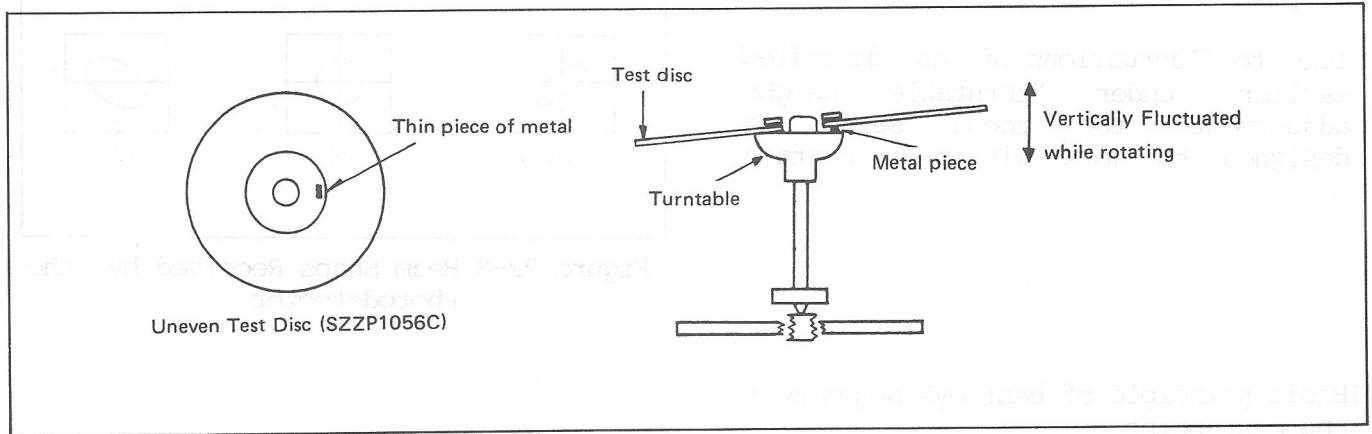


Figure 22-5

(2) RF waveform when properly aligned

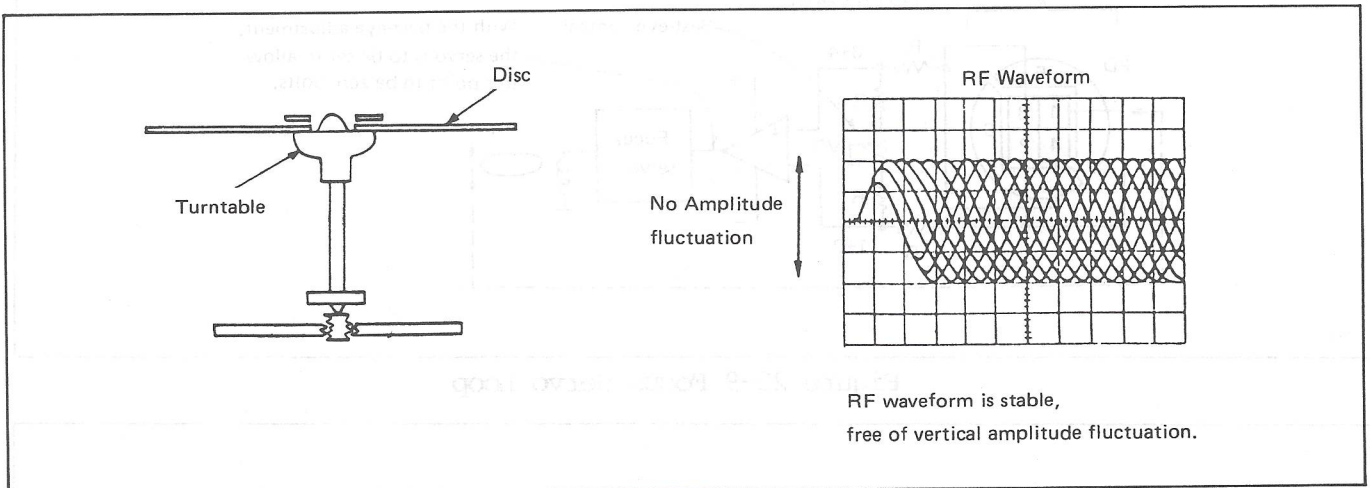


Figure 22-6

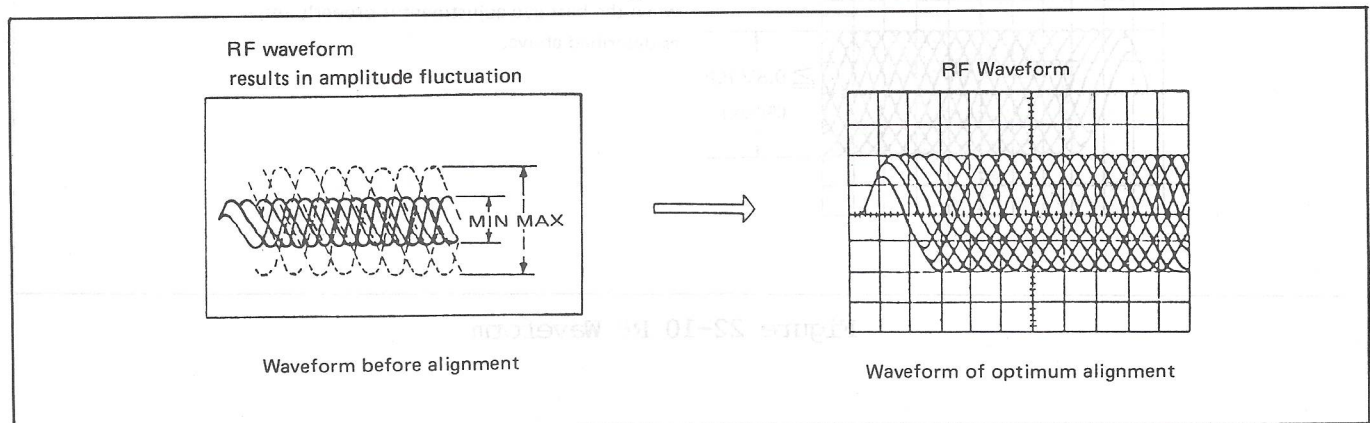


Figure 22-7

c) Basic principle of adjustment

Due to distance variations between the disc surface and the lens caused by uneven discs, the laser beam covering a given pit varies in diameter. When

the lens is close to the disc the diameter is large and covers adjacent pits. The signal from adjacent pits causes interference and the RF becomes distorted.

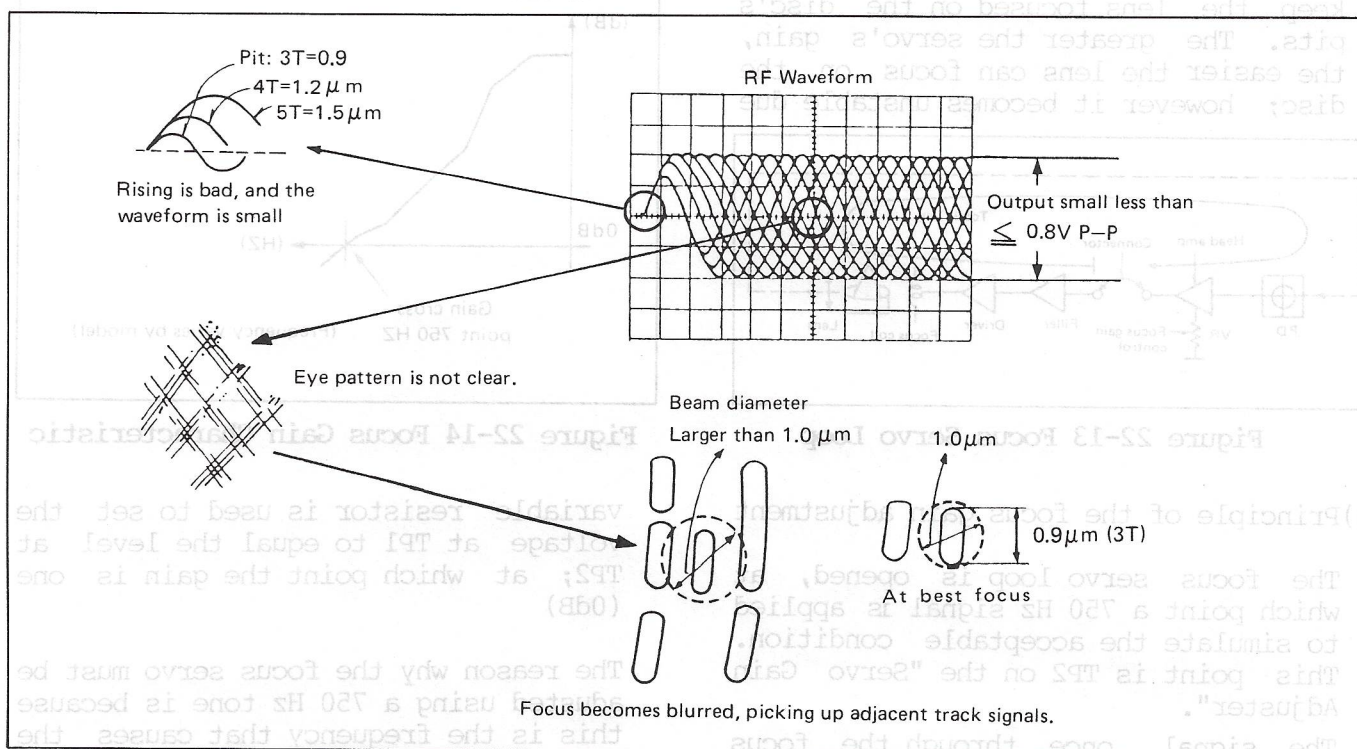


Figure 22-11

d) Adjustment method

It is required to adjust so that the RF waveform is maximum and the eye pattern appears clearly.

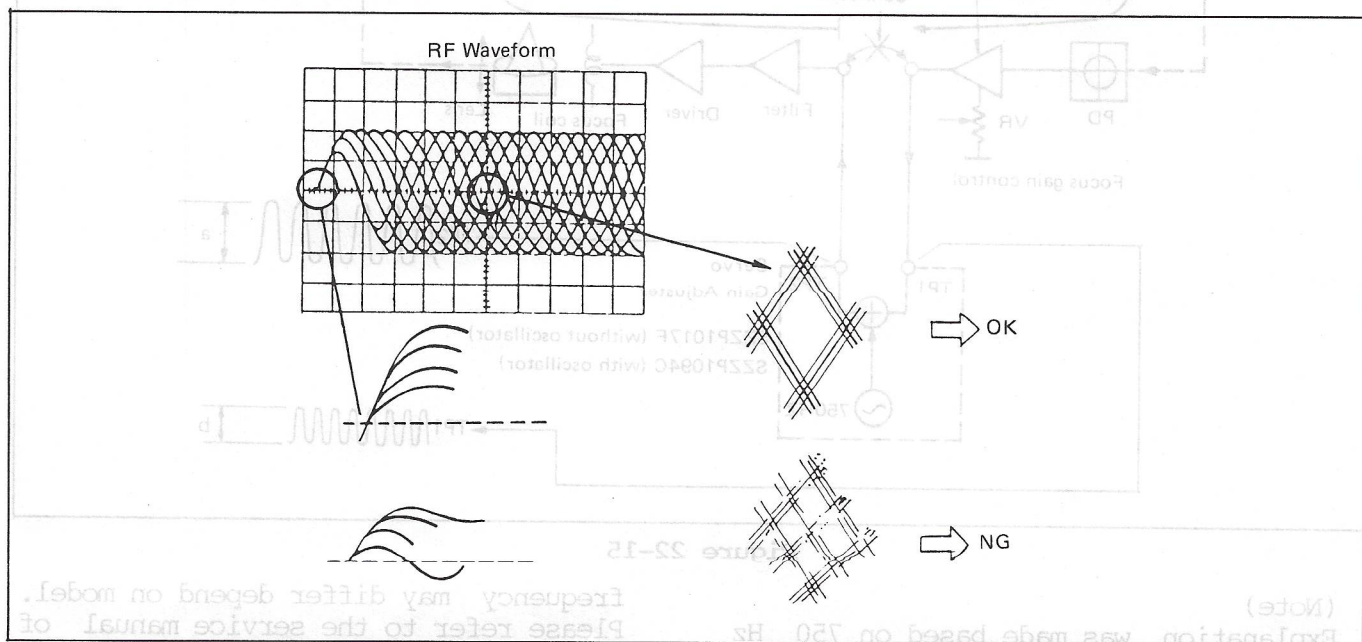


Figure 22-12

B. Focus Gain Adjustment

a) Purpose

Regardless of disc warpage ($\pm 0.4\text{mm}$) due to manufacturing tolerances, the focus servo must operate to constantly keep the lens focused on the disc's pits. The greater the servo's gain, the easier the lens can focus on the disc; however it becomes unstable due

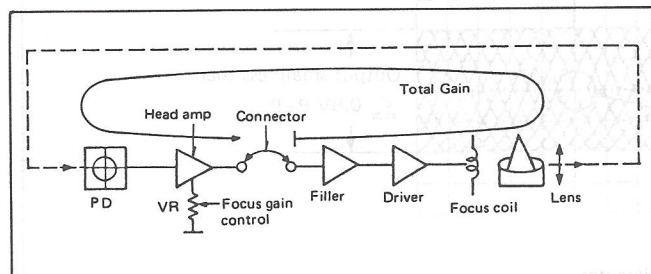


Figure 22-13 Focus Servo Loop

b) Principle of the focus gain adjustment

The focus servo loop is opened, at which point a 750 Hz signal is applied to simulate the acceptable condition. This point is TP2 on the "Servo Gain Adjuster".

The signal, once through the focus servo, appears at TP1. The focus gain

to unwanted signals. The gain is therefore set to unity ($0\text{ dB}=1$, 750Hz) at which point stability is acceptable and the gain is adequate.

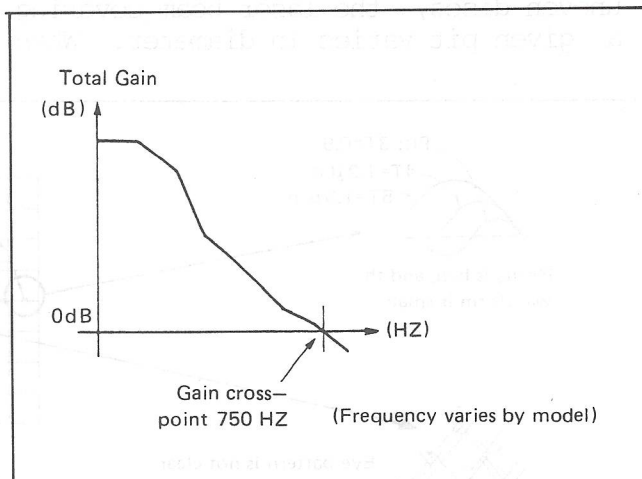


Figure 22-14 Focus Gain Characteristic

variable resistor is used to set the voltage at TP1 to equal the level at TP2; at which point the gain is one (0 dB)

The reason why the focus servo must be adjusted using a 750 Hz tone is because this is the frequency that causes the gain to become unity ($0\text{ dB}=1$)

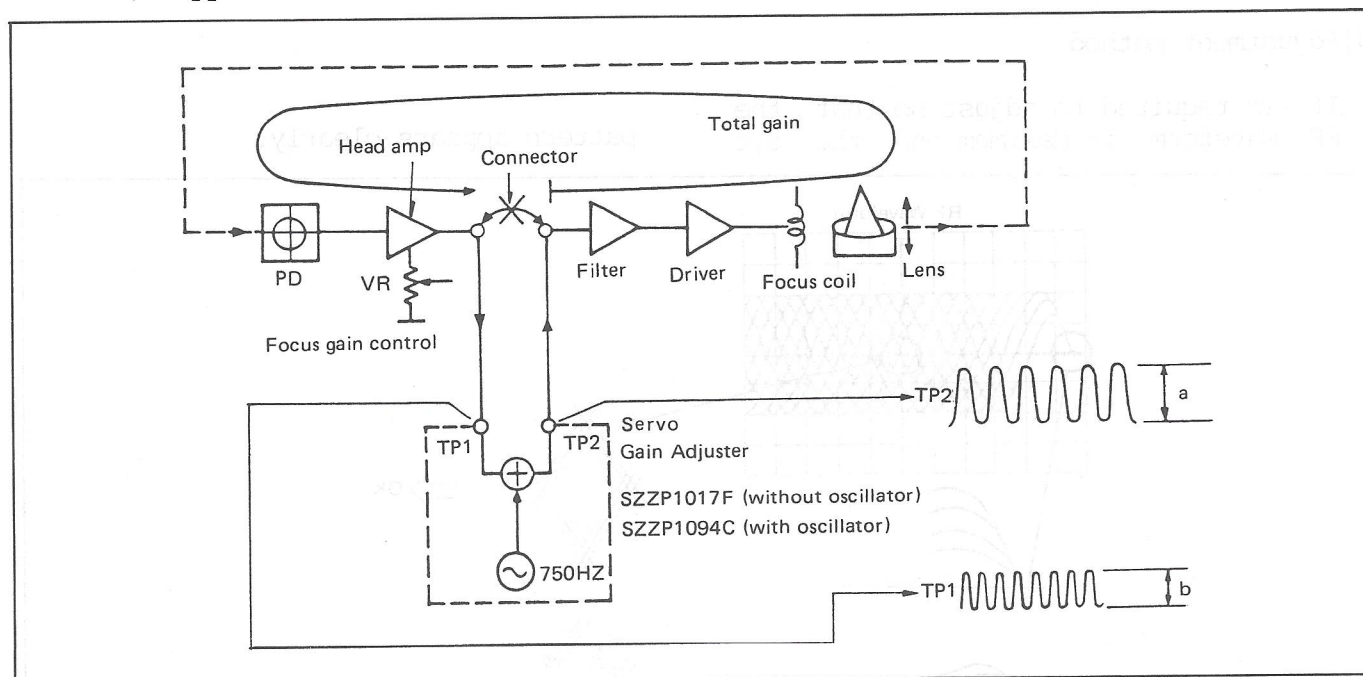


Figure 22-15

(Note)

Explanation was made based on 750 Hz gain cross-point frequency. This

frequency may differ depend on model. Please refer to the service manual of the corresponding model for frequency.

D. Focus Offset Adjustment

a) Purpose

The CD player must be able to compensate for manufacturing imperfections of disc and dust or any other contaminants on the signal side.

As the pickup traces the track, if dust or disc imperfections are present, the focusing depth changes.

Without proper focus the tracking servo is affected as well, causing the pickup to move to a different track.

When the lens detects imperfections as mentions above, the focus servo is enabled in order to maintain proper focusing within the operating range of the servo.

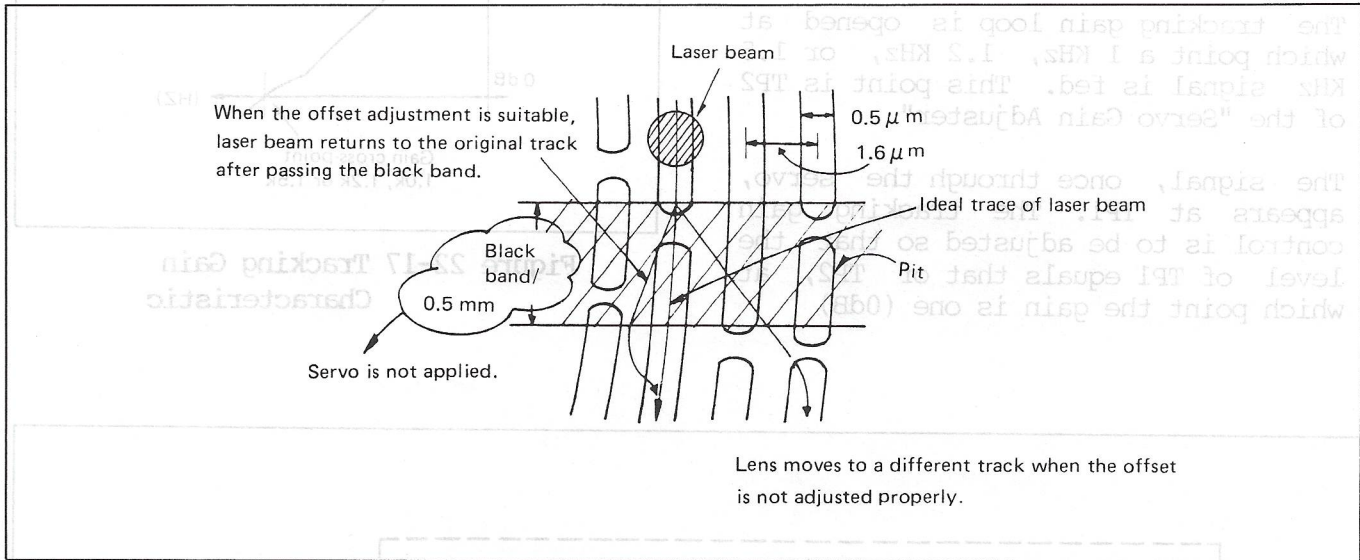


Figure 22-18

b) Test disc, RF waveform and FE signal
A black band disc is used to simulate disc imperfection or dust particles,

causing the RF to appear as in figure 22-21 (a) and (b).

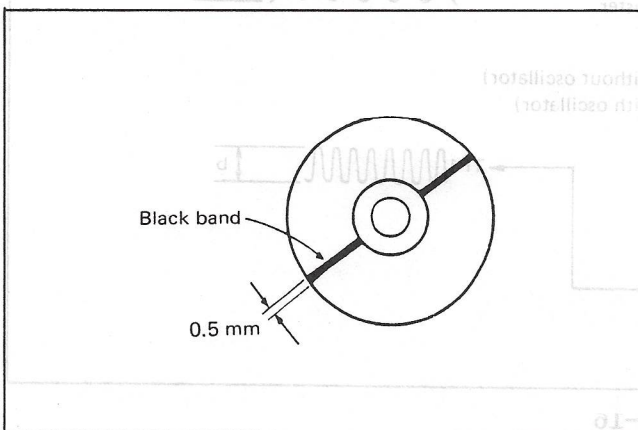


Figure 22-19 Test Disc (SZZP1057C)

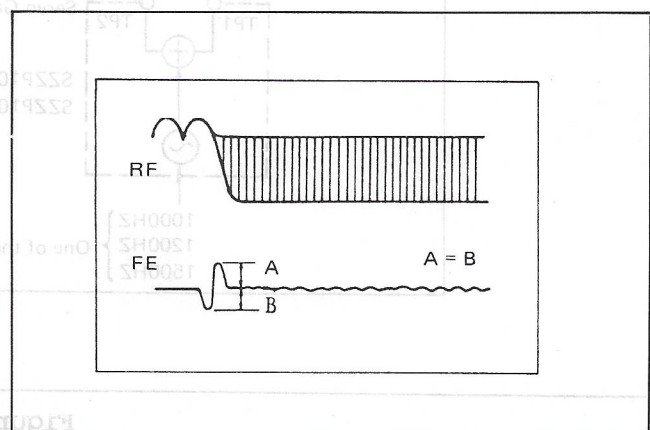


Figure 22-20 RF Waveform When Adjusted

C. Tracking Gain Adjustment

a) Purpose

During disc manufacturing, disc eccentricity results due to tooling tolerances. Eccentricity also results if the tracking servo is designed to overcome these tolerances and keep the lens on track.

b) Principle of the tracking gain adjustment

The tracking gain loop is opened at which point a 1 KHz, 1.2 KHz, or 1.5 KHz signal is fed. This point is TP2 of the "Servo Gain Adjuster".

The signal, once through the servo, appears at TP1. The tracking gain control is to be adjusted so that the level of TP1 equals that of TP2, at which point the gain is one (0dB)

The reason why the focus servo to be adjusted using a 1 KHz, 1.2 KHz or 1.5 KHz tone is because these are the frequencies that cause the gain to become unity (0dB=1).

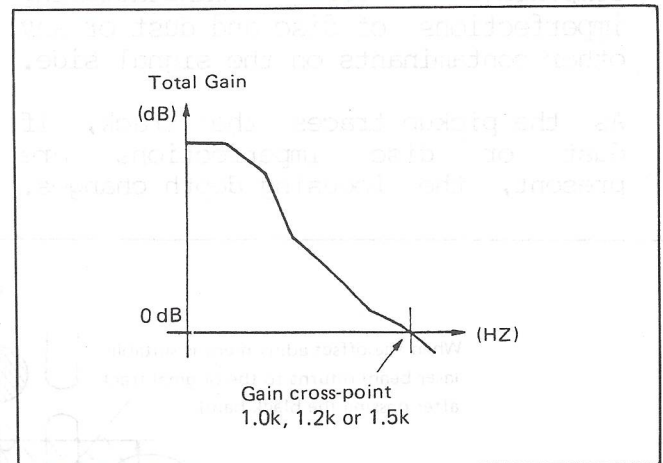


Figure 22-17 Tracking Gain Characteristic

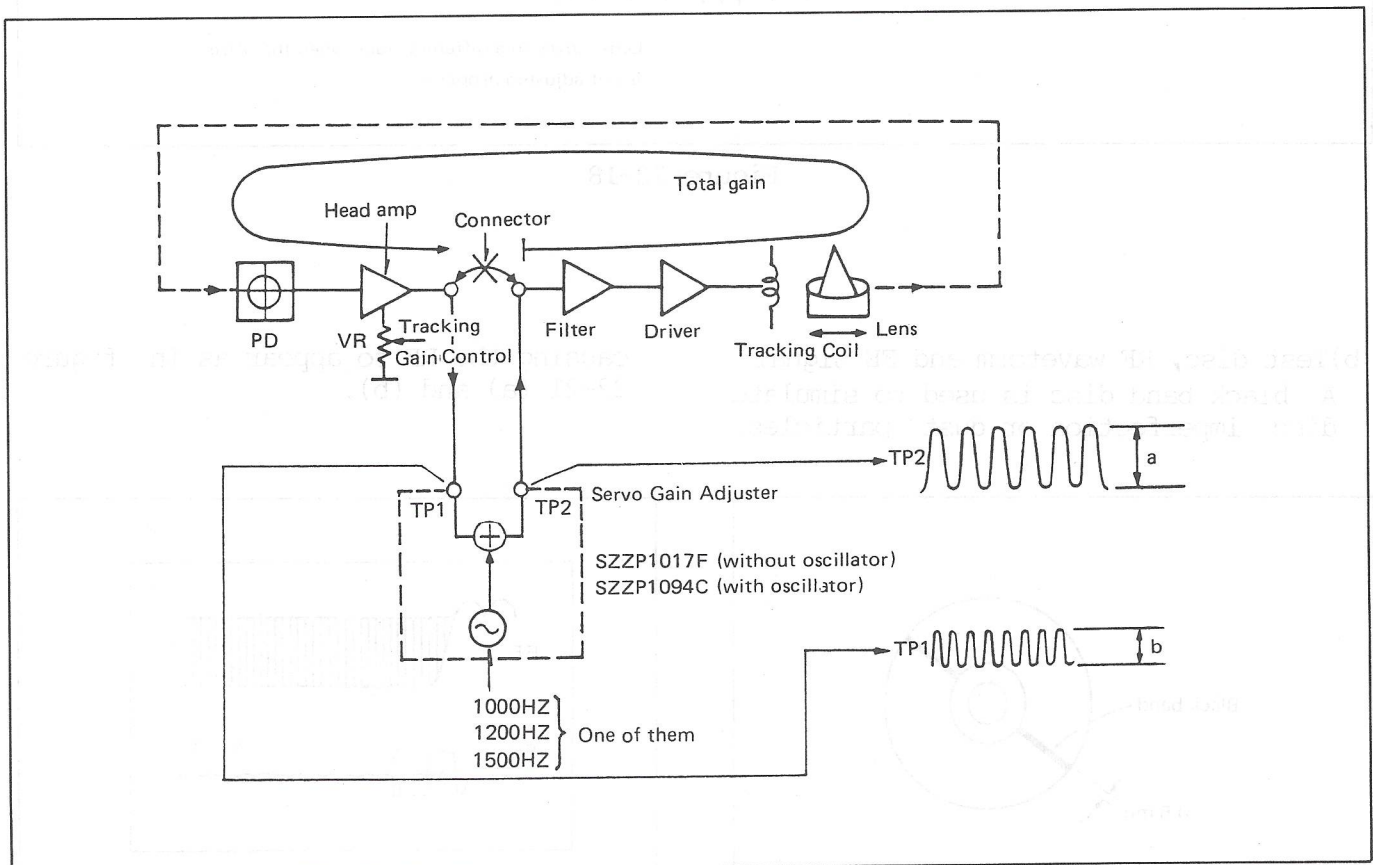


Figure 22-16

c) Focus offset adjustment method

It should be adjusted so that the RF waveform dip is minimized or eliminated and the size of FE (a) and (b) is at minimum or balanced.

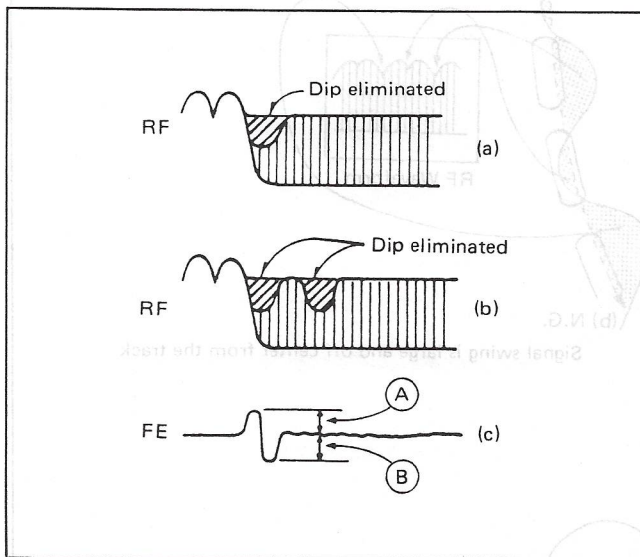


Figure 22-21

c) Tracking offset adjustment method

The RF waveform must be adjusted such that the dip's is are minimized or eliminated as indicated below in figure 22-22 (a) and (b).

The TE waveform must be adjusted such that (a) and (b) are equal, as indicated in figure 22-22 (c) below.

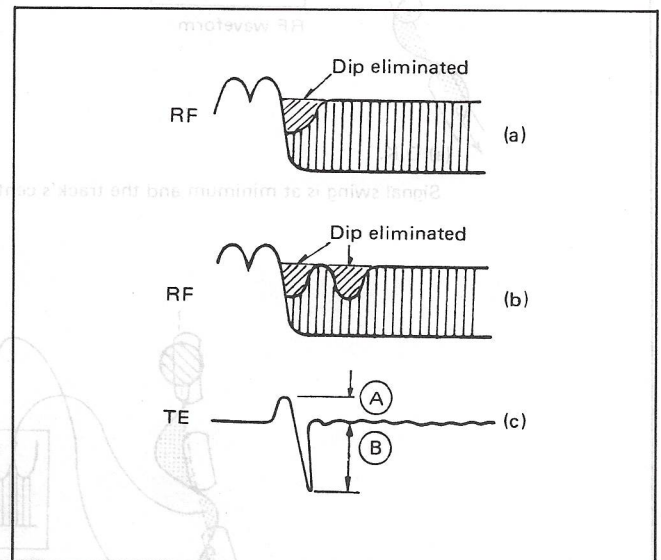


Figure 22-22

E. Tracking Offset Adjustment

a) Purpose

As with the focus offset adjustment, the tracking offset adjustment is necessary to set reference point of the tracking servo operating range.

The only difference is that the tracking servo compensate for horizontal control as opposed to vertical control by the focus servo.

b) Test disc, RF waveform and TE signal

The test disc, RF waveform, and TE (Tracking Error) signal waveform are the same as described in the focus offset adjustment. The only difference is the TE signal.

F. Tracking Offset Balance Adjustment

a) Purpose

During playback operation, the laser beam must follow the track exactly at its center. If it is not, proper tracking can not be accomplished, and skipping may occur during playback of the disc which has manufacturing imperfections and dust or any other contaminants on the signal side.

b) Basic principle of adjustment

A 1 KHz signal is applied to the tracking servo circuit to intentionally cause the beam to oscillate from side to side of the being traced (see figure 22-23).

This is done to simulate an error signal so that the tracking servo can be adjusted for minimum gain (minimum swing) at 1 KHz.

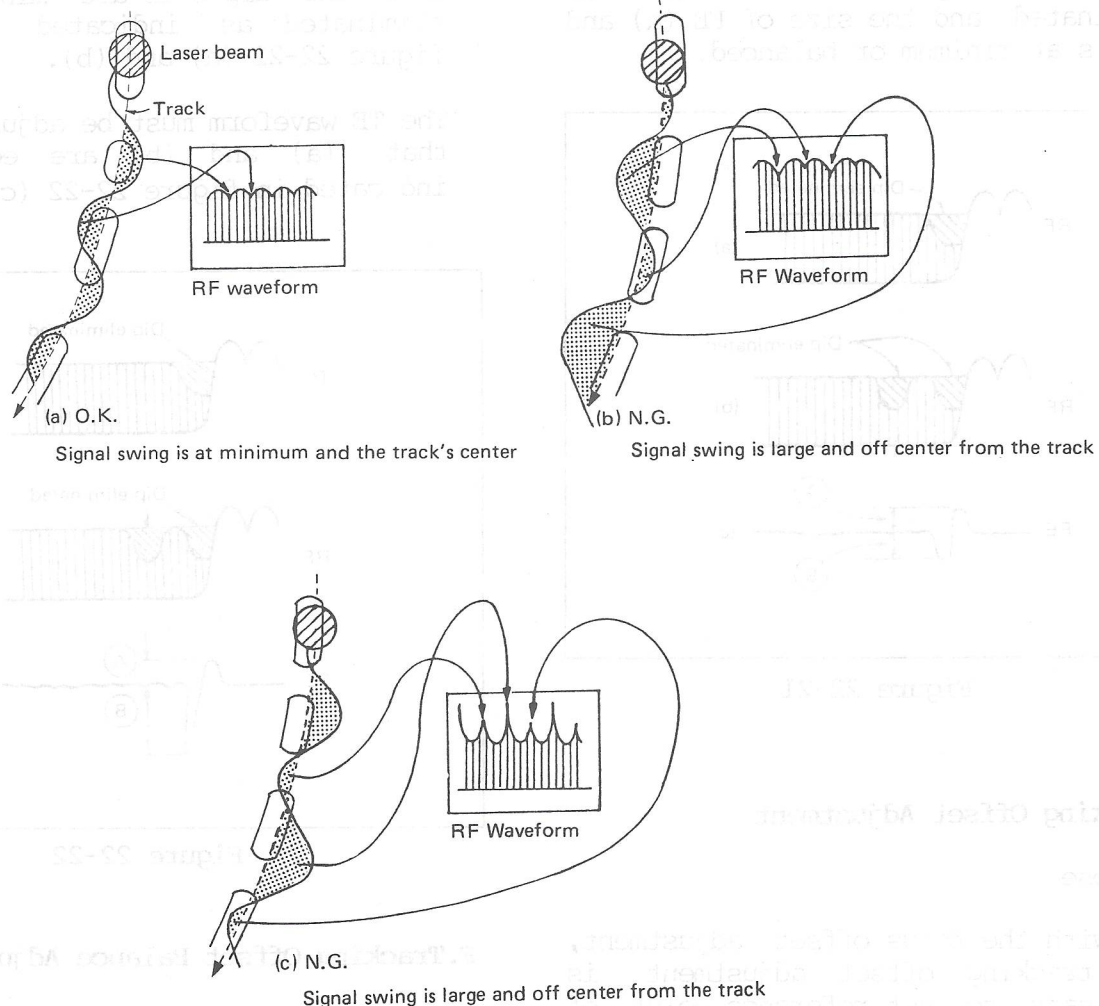


Figure 22-23 Beam Movement with Error Signal